



MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 23

Level 2 — Intermediate · Number Sense III

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Number Sense III — Factors, Multiples & Primes
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	23 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 23 introduces the building blocks of numbers: factors, multiples, common factors and common multiples, and a first friendly meeting with prime numbers. The bell-and-buzzer question is the classic common-multiple application, and the 'Who am I?' riddles teach systematic checking — list the candidates, test each clue.

Learning Objectives

- Find factors and multiples of numbers up to 50.
- Identify common factors and the first common multiples of two numbers.
- Recognise prime numbers and the special role of 1, 2 and 0.
- Apply divisibility by 2, 5 and 10 from the last digit.
- Solve real-life common-multiple problems (bells, teams) and number riddles.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

**SECTION A · Multiple Choice Questions · 20 Questions**

Q1. Which of these is a FACTOR of 24?

- (A) 5 (B) 7
(C) 8 (D) 9

Q2. Which of these is a MULTIPLE of 9?

- (A) 35 (B) 54
(C) 65 (D) 75

Q3. How many factors does the number **12** have in all?

- (A) 4 (B) 5
(C) 6 (D) 8

Q4. What is the SMALLEST multiple of any number?

- (A) 0 (B) 1
(C) The number itself (D) Twice the number

Q5. Which number is a factor of EVERY number?

- (A) 0 (B) 1
(C) 2 (D) 10

Q6. The 7th multiple of 8 is:

- (A) 48 (B) 54
(C) 56 (D) 64

Q7. Which number is divisible by 2 AND by 5?

- (A) 25 (B) 32
(C) 40 (D) 54

Q8. I am a number between 20 and 30. I am a multiple of both 3 and 4. Who am I?

- (A) 21 (B) 24
(C) 27 (D) 28

Q9. Which of these numbers has an ODD number of factors?

- (A) 10 (B) 15
(C) 16 (D) 21

Q10. The COMMON factors of 18 and 24 are:

- (A) 1, 2, 3, 6 (B) 1, 2, 4, 6
(C) 2, 3, 4, 6 (D) 1, 3, 6, 9

Q11. Which is a PRIME number (exactly two factors: 1 and itself)?

- (A) 9 (B) 15
(C) 13 (D) 21



Q12. The smallest EVEN prime number is:

- (A) 0 (B) 1
(C) 2 (D) 4

Q13. A number is divisible by 10 if it ends in:

- (A) 5 (B) 0
(C) Any even digit (D) 1

Q14. Which list shows ONLY multiples of 6?

- (A) 6, 12, 16, 24 (B) 6, 12, 18, 24
(C) 6, 14, 18, 22 (D) 12, 18, 26, 30

Q15. 36 students must be split into EQUAL teams. Which team size is NOT possible?

- (A) 4 (B) 6
(C) 9 (D) 8

Q16. What is the GREATEST factor of 45 other than 45 itself?

- (A) 9 (B) 15
(C) 5 (D) 20

Q17. The first THREE common multiples of 2 and 3 are:

- (A) 2, 3, 6 (B) 6, 12, 18
(C) 6, 9, 12 (D) 3, 6, 9

Q18. A bell rings every 4 minutes and a buzzer sounds every 6 minutes. They sound TOGETHER now. After how many minutes will they sound together AGAIN?

- (A) 10 (B) 12
(C) 18 (D) 24

Q19. Which statement is FALSE?

- (A) Every number is a factor of itself (B) Every number is a multiple of 1
(C) The number of multiples of a number is endless (D) 0 is a factor of every number

Q20. I am a 2-digit number. I am a multiple of 7, and my two digits add up to 6. Who am I?

- (A) 42 (B) 35
(C) 21 (D) 63



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	B	C	C	B	C	C	B	C	A
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	C	B	B	D	B	B	B	D	A

Step-by-Step Solutions

Q1. Correct answer: (C)

$24 \div 8 = 3$ exactly, so **8** is a factor of 24. ($24 \div 5, \div 7, \div 9$ all leave remainders.)

Q2. Correct answer: (B)

$54 = 9 \times 6$, so it is a multiple of 9.

Q3. Correct answer: (C)

Factors of 12: 1, 2, 3, 4, 6, 12 — that is **6 factors**.

Q4. Correct answer: (C)

The smallest (first) multiple of a number is **the number itself** (e.g., multiples of 7: 7, 14, 21 ...).

Q5. Correct answer: (B)

1 divides every number exactly, so 1 is a factor of every number.

Q6. Correct answer: (C)

$7 \times 8 = 56$.

Q7. Correct answer: (C)

A number divisible by both 2 and 5 must end in 0. **40** ends in 0.

Q8. Correct answer: (B)

Multiples of 3 and 4 are multiples of 12: 12, **24**, 36 ... Only 24 lies between 20 and 30.

Q9. Correct answer: (C)

16 has factors 1, 2, 4, 8, 16 — five factors (odd). Square numbers always have an odd number of factors because one factor (here 4) pairs with itself.

Q10. Correct answer: (A)

Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Common: **1, 2, 3, 6**.

Q11. Correct answer: (C)

13 can only be divided by 1 and 13. ($9 = 3 \times 3$, $15 = 3 \times 5$, $21 = 3 \times 7$.)

Q12. Correct answer: (C)

2 is the only even prime — every other even number is divisible by 2 and so has extra factors.

Q13. Correct answer: (B)

Multiples of 10 always end in **0** (10, 20, 30 ...).

Q14. Correct answer: (B)

6, 12, 18, 24 are all in the table of 6. (16, 14, 22 and 26 are not.)

Q15. Correct answer: (D)

Teams must be factors of 36. $36 \div 4 = 9 \checkmark$, $\div 6 = 6 \checkmark$, $\div 9 = 4 \checkmark$, but $36 \div 8$ leaves a remainder **X**.

**Q16. Correct answer: (B)**

Factors of 45: 1, 3, 5, 9, 15, 45. The greatest below 45 is **15**.

Q17. Correct answer: (B)

Common multiples of 2 and 3 are the multiples of 6: **6, 12, 18**.

Q18. Correct answer: (B)

They meet at common multiples of 4 and 6. The first is **12 minutes** (4: 4,8,12 ...; 6: 6,12 ...).

Q19. Correct answer: (D)

'0 is a factor of every number' is **FALSE** — division by 0 is not allowed, so 0 is never a factor. The other three statements are true.

Q20. Correct answer: (A)

2-digit multiples of 7: 14,21,28,35,42,49,56,63,70,77,84,91,98. Digit sums: 21→3, 42→**6** ✓, 63→9, 35→8. The answer is 42.

